MATHEMATICS

DIFFERENTIAL EQUATION

THEORY SHEET

ENGLISH

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1.1 INTRODUCTION

An equation involving independent and dependent variables and the derivatives of the dependent variables is called a **differential equation**. There are two kinds of differential equation:

1 Ordinary Differential Equation : If the dependent variables depend on one independent variable x, then the differential equation is said to be ordinary.

for example $\frac{dy}{dx} + \frac{dz}{dx} = y + z$, $\frac{dy}{dx} + xy = \sin x$

2. Partial differential equation : If the dependent variables depend on two or more independent variables, then it is known as partial differential equation

for example $y^2\frac{\partial z}{\partial x} + y\frac{\partial^2 z}{\partial y} = ax$, $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$

Finding the unknown function is called **SOLVING OR INTEGRATING** the differential equation. The solution of the differential equation is also called its **PRIMITIVE**

$\frac{dy}{dx} = 2x$

$y = x^2 + c$ This is general solution

General Solution is also called Complete Integral/Complete Primitive.

A solution obtainable from the general solution by giving particular values to the constants is called a **Particular Solution**.

Ex.1 Find equation of curve passing through (2, 5) and satisfying the differential equation $\frac{dy}{dx} = 2x$

Sol $\frac{dy}{dx} = 2x$

$y = x^2 + c$

This curve passes through (2, 5) so $c = 1$

$y = x^2 + 1$

1.2 ORDER AND DEGREE OF A DIFFERENTIAL EQUATION

Order :

It is the order of the highest order derivative appearing in a differential equation.

Degree :

After expressing the differential equation as polynomial of derivatives, the highest power of highest order derivative is its degree.

Ex. 1 $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = y^2 + \sin x$

Ans Degree = 1. Order = 2

Ex. 2 $\frac{d^2y}{dx^2} = \left(1 + \frac{dy}{dx}\right)^{1/3}$

Ans Degree = 3. Order = 2

Ex. 3 $\sin^{-1}\left(\frac{dy}{dx}\right) = x + 2$

Ans Degree = 1. Order = 1

Ex. 4 $\sin^{-1}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2} + x$

Ans Degree = not defined. Order = 2

1.3 FORMATION OF A DIFFERENTIAL EQUATION

Independent arbitrary constants.

The arbitrary constants in the solution of a differential equation are said to be independent, when it is impossible to deduce from the solution an equivalent relation containing fewer arbitrary constants.

Ex.1 Find number of independent arbitrary constants for following family of curves?

1. $y = mx + c$

Ans 2

2. $ax + by + c = 0$

Ans 2

Sol $\frac{ax}{c} + \frac{b}{c}y + 1 = 0$

$a_1x + b_1y + 1 = 0$

3. $y = c_1e^{x+c_2}$

Ans 1

Sol $y = \left(c_1e^{c_2}\right)e^x$

$y = Ae^x$.

4. $y^2 = 4A(x + A)$

Ans 1

Sol $y^2 = 4Ax + 4A^2$

5. $y^2 = 4A(x + B)$

Ans 2

Sol $y^2 = 4Ax + 4AB$

How to form differential equation for a family of curves

(a) Differentiate the given equation w.r.t the independent variable (say x) as many times as the number of independent arbitrary constants in it.

(b) Eliminate the arbitrary constants.

(c) The eliminant is the required differential equation.

Ex.2 Form differential Equation for following family of curves

(i) $y = 3x + c$ (ii) $y = mx$ (iii) $x^2 + y^2 = r^2$

(iv) $y = ce^x$ (v) $y = ax^2 + bx + c$ (vi) $y = Ae^{2x} + Be^{-2x} + C$

Ans. (i) $\frac{dy}{dx} = 3$ (ii) $\frac{dy}{dx} = \frac{y}{x}$ (iii) $\frac{dy}{dx} = \frac{-x}{y}$

(iv) $\frac{dy}{dx} = y$ (v) $\frac{d^3y}{dx^3} = 0$ (vi) $\frac{d^3y}{dx^3} = 4\frac{dy}{dx}$

Sol. (i) $y = 3x + c \Rightarrow \frac{dy}{dx} = 3$

(ii) $y = mx \Rightarrow \frac{dy}{dx} = m \Rightarrow \frac{y}{x} = \frac{dy}{dx}$

(iii) $x^2 + y^2 = r^2 \Rightarrow 2xdx + 2ydy = 0$

$\Rightarrow xdx + ydy = 0$

(iv) $y = ce^x \Rightarrow \frac{dy}{dx} = ce^x = y$

(v) $y = ax^2 + bx + c \Rightarrow \frac{dy}{dx} = 2ax + b$

$\Rightarrow \frac{d^2y}{dx^2} = 2a \Rightarrow \frac{d^3y}{dx^3} = 0$

(vi) $y = ae^{2x} + be^{-2x} + c \Rightarrow \frac{dy}{dx} = 2ae^{2x} - 2be^{-2x}$

$\Rightarrow \frac{d^2y}{dx^2} = 4ae^{2x} + 4be^{-2x}$

$\Rightarrow \frac{d^3y}{dx^3} = 8ae^{2x} - 8be^{-2x} = 4\frac{dy}{dx}$

Questions on Formation of Differential Equation

1. Form the differential equation of family of lines concurrent at the origin

Ans $xdy - ydx = 0$

Sol. $y = mx \Rightarrow \frac{dy}{dx} = m = \frac{y}{x}$

2. Form Differential equation of all concentric circle at the origin

Ans $xdx + ydy = 0$

Sol. $x^2 + y^2 = r^2 \Rightarrow 2xdx + 2ydy = 0 \Rightarrow xdx + ydy = 0$

3. Form differential equation of all circles touching the x axis at the origin and centre on y-axis

Ans. $(x^2 - y^2)\frac{dy}{dx} = 2xy$

Sol $x^2 + (y - a)^2 = a^2$

$x^2 + y^2 - 2ay = 0$ ....(1)

differentiating , $2x + 2y\frac{dy}{dx} = 2a\frac{dy}{dx}$ or $x + y\frac{dy}{dx} = a\frac{dy}{dx}$

substituting the value of 'a' in (1)

$(x^2 - y^2)\dfrac{dy}{dx} = 2xy$ (order is one again and degree 1)

4. Form the differential equation of the family of parabolas with focus at the origin and axis of symmetry along the x-axis.

Ans. $y^2 = y^2\left(\dfrac{dy}{dx}\right)^2 + 2xy\dfrac{dy}{dx}$

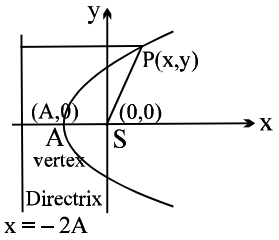
Sol equation of the parabola is

$(x^2 + y^2) = (x + 2A)^2$

$y^2 = 4A(A + x)$

$2y\dfrac{dy}{dx} = 4A$

$\therefore y\dfrac{dy}{dx} = 2A$



Hence $y^2 = \left(y\dfrac{dy}{dx}\right)^2 + 2y\dfrac{dy}{dx}.x$ or $y^2 = y^2\left(\dfrac{dy}{dx}\right)^2 + 2xy\dfrac{dy}{dx}$ order 1, degree 2

5. Form the differential equation of family of lines situated at a constant distance p from the origin.

Ans. $\left(y - x\dfrac{dy}{dx}\right)^2 = p^2\left(1 + \left(\dfrac{dy}{dx}\right)^2\right)$

Sol. $y = mx + p\sqrt{1 + m^2}$ (Tangents to the circle $x^2 + y^2 = p^2$)

where $m = \dfrac{dy}{dx}$

Note that order is 1 and degree is 2.

Method II

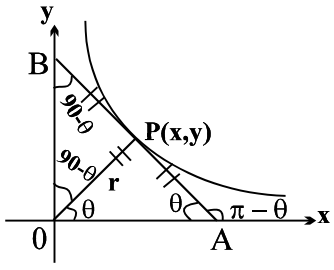
Consider the line $x\cos\alpha + y\sin\alpha = p$ where p is a constant and α varies. Diffrentiate w.r.t. x i.e. $\cos\alpha + y'\sin\alpha = 0 \Rightarrow y' = -\cot\alpha$. Put in the given line]

6. Form the differential of the system of rectangular hyperbola $xy = c^2$

Ans. $xdy + ydx = 0$

Sol diff., $xdy + ydx = 0$

$\Rightarrow \dfrac{dy}{dx} = -\dfrac{y}{x}$



7. Form Differential equation for family of curves $y^2 = 2c(x + \sqrt{c})$

Ans $(y^2 - 2xyy')^2 = 4(yy')^3$

Sol. $y^2 = 2c\left(x + \sqrt{c}\right) \Rightarrow 2yy' = 2c \Rightarrow c = yy'$

$\Rightarrow y^2 = 2yy'\left(x + \sqrt{yy'}\right) \Rightarrow y^2 - 2yy'x = 2\left(yy'\right)^{\frac{3}{2}}$

$\Rightarrow (y^2 - 2yy'x)^2 = 4(yy')^3$

8. Form the differential equation of the family of concentric ellipse $ax^2 + by^2 = 1$ with principal axes along the co-ordinate axes.

Ans $x(y')^2 + xyy'' = yy'$

Sol. $2ax + 2byy_1 = 0 \Rightarrow ax + byy_1 = 0$ ...(1) ⎤ equate a/b to get a D.E. of

again diff. $a + b(yy_2 + y_1^2) = 0$ ....(2) ⎦ order 2 and degree 1

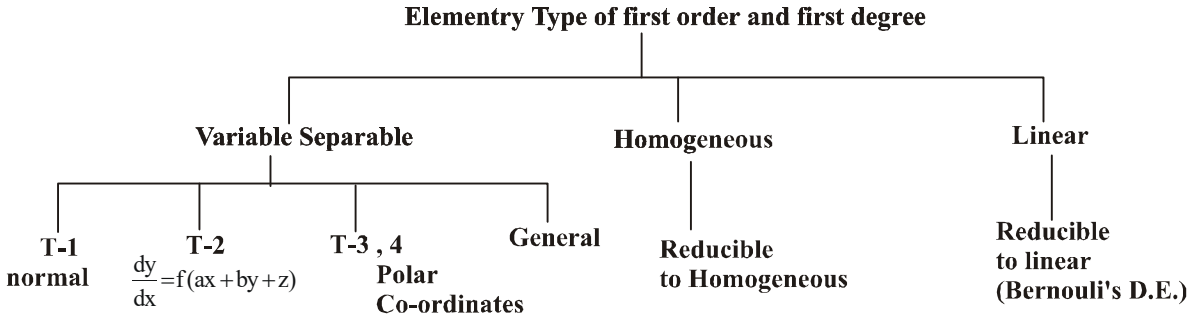
9. Form the differential equation for family of parabolas with axis of symmetry parallel to y axis ;

Ans. $\frac{d^3y}{dx^3} = 0$

Sol $y = ax^2 + bx + c;$

diff. thrice $\Rightarrow \frac{d^3y}{dx^3} = 0.$

2. ELEMENTARY TYPES OF FIRST ORDER & FIRST DEGREE DIFFERENTIAL EQUATIONS.



2.1 VARIABLE SEPARABLE - TYPE 1

If the differential equation can be put in the form, $f(x)\,dx = \phi(y)\,dy$ we say that variables are separable and solution can be obtained by integrating each side separately.

A general solution of this will be $\int f(x)\,dx = \int \phi(y)\,dy + c$, where c is an arbitrary constant

Questions

1. $\ln \frac{dy}{dx} = ax + by$

Ans. $\frac{e^{ax}}{a} = \frac{-e^{-by}}{b} + c$

Sol $\frac{dy}{dx} = e^{ax+by} = e^{ax} \cdot e^{by}$ or $\int e^{-by}\,dy = \int e^{ax}\,dx$

2. $x(y^2 + 1)\,dx + y(x^2 + 1)\,dy = 0$

Ans. $(x^2+1)(y^2+1) = C$

Sol. $x(y^2 + 1)dx + y(x^2 + 1)dy = 0$

$\Rightarrow \frac{x}{x^2+1}dx + \frac{y}{y^2+1}dy = 0$

$\Rightarrow \frac{1}{2}\ell n\left(x^2+1\right) + \frac{1}{2}\ell n\left(y^2+1\right) = \ell nc$

$\Rightarrow (x^2 + 1)(y^2 + 1) = c$

3 $y' \sin x = y \ln y$; $y\big|_{x=\frac{\pi}{2}} = e$

Ans. $y = e^{\text{cosecx} - \cot x}$

Sol $\int \frac{dy}{\ln y \cdot y} = \int \text{cosec}\, x\, dx \quad \Rightarrow \quad \ln(\ln y) = \ln(\text{cosec}x - \cot x) + C$

4. $(dy/dx) = e^{x-y} + x^2 . e^{-y}$.

Ans $e^y = e^x + \frac{x^3}{3} + c$

Sol. $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y} = e^{-y}\left(e^x + x^2\right)$

$\Rightarrow e^y dy = (e^x + x^2)e^x$

$\Rightarrow e^y = e^x + \frac{x^3}{3} + C$

5. The equation of the curve passing through (3, 9) which satisfies differential equation $dy/dx = x + 1/x^2$ is

Ans. $6xy = 3x^3 + 29x - 6$

Sol. **(C)** The given differential equation has variable separable so integrating, we have $y = x^2/2 - 1/x + C$. This will pass through (3, 9) if $9 = 9/2 - 1/3 + C \Rightarrow C = 29/6$.

Hence the required equation is $6xy = 3x^3 + 29x - 6$.

6. Find the foci of the conic passing through the point (1, 0) and satisfying the differential equation $(1+ y^2)\, dx - xydy = 0$.

Ans $x^2 - y^2 = 1$; foci $= \left(\pm\sqrt{2}, 0\right)$

Sol. $(1 + y^2)dx = xydy \Rightarrow \frac{dx}{x} = \frac{ydy}{1+y^2}$

$\Rightarrow \ell n|x| = \frac{1}{2}\ell n\left|y^2 + 1\right| + \ell nC$

$\Rightarrow x^2 = c(y^2 + 1)$

Passing through (1, 0) $\Rightarrow c = 1 \Rightarrow x^2 - y^2 = 1$

Sol eccentricity $= \sqrt{2}$; conic: $x^2 - y^2 = 1$ (Rectangular Hyperbola)

foci $= \left(\pm\sqrt{2}, 0\right)$

7. Find a particular solution of the differential equation $(1 + e^x)y\frac{dy}{dx} = e^x$. Satisfying the initial condition $y(0) = 1$.

Ans. $y = \sqrt{1 + \ln\left(\frac{1+e^x}{2}\right)^2}$

Sol. $ydy = \frac{e^x}{1+e^x}dx \Rightarrow \frac{y^2}{2} = \ell n\left(e^x + 1\right) + c$

Passes through (0, 1)

$$c = \frac{1}{2} - \ell n2$$

$$\frac{y^2}{2} = \ell n\left(e^x + 1\right) + \frac{1}{2} - \ell n2 = \ell n\left(\frac{e^x + 1}{2}\right) + \frac{1}{2}$$

$$y = \sqrt{1 + 2\ell n\left(\frac{e^x + 1}{2}\right)}$$

Only (+) sign as y(0) = 1

8. Find the curve passing through the point (0, –2) such that the slope of the tangent at any of its points is equal to the ordinate of that point augumented by a factor of 3.

Ans $y = -2e^{3x}$

Sol. Given $\frac{dy}{dx} = 3y \Rightarrow \frac{dy}{y} = 3dx$

$\Rightarrow \ell n|y| = 3x + \ell n\, C$

Pass through (0, –2)

$\ell nC = \ell n2$

$$\Rightarrow \ell n\left|y\right| = 3x + \ell n2 \Rightarrow \ell n\left|\frac{y}{2}\right| = 3x$$

$\Rightarrow y = 2e^{3x}$ or $-2e^{3x}$

But (0, –2) satisfies.

Hence, $y = -2e^{3x}$.

9. Find the nature of the curve not passing through the origin for which the length of the normal (tangent) at the point P is equal the radius vector of the point P.

Ans. Circle or rectangular hyperbola and
straight line or rectangular hyperbola

Sol. [Equation of normal : $Y - y = -\frac{1}{m}(X - x)$

$$OP^2 = GP^2$$

$$x^2 + y^2 = m^2 y^2 + y^2$$

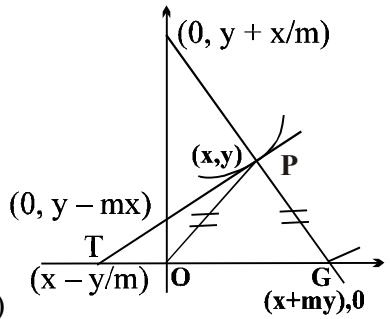
$$\therefore \quad m = \pm \frac{x}{y}$$

with +ve, $xdx - ydy = 0$

$x^2 - y^2 = C$ (Rectangular Hyperbola)

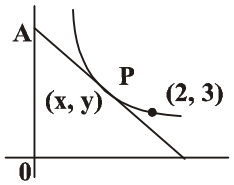
with –ve $x^2 + y^2 = C$ (circle)

|||ly we can do for tangent where $Y - y = m(X - x)$



10. Find the curve for which the segment of the tangent contained between the co-ordinate axes is bisected by the point . Curve passes through (2, 3).

(ii) In the figure, find the curve if passes through (2, 0) and $l(PA) = 2$.



Ans: (i) $xy = 6$

(ii) $y = 2\ \ln\left|\frac{2-\sqrt{4-x^2}}{x}\right| + \sqrt{4-x^2}$ or

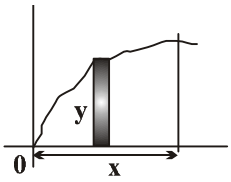
$y = -2\ \ell n\left|\frac{2-\sqrt{4-x^2}}{x}\right| - \sqrt{4-x^2}$

11. Find the curve $y = f(x)$ $(f(x) \geq 0$ and $f(0) = 0)$ bounding a curvilinear trapezoid with the base $[0, x]$ if area bounded by curve, coordinate axes & var. ordinate. Area $\propto (f(x))^{n+1}$ and $f(1) = 1$

Ans. $f(x) = x^{1/n}$

Sol

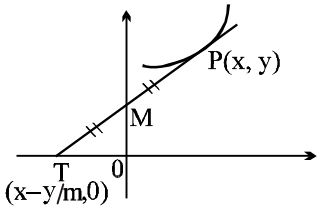
$\int_0^x y\,dx = K.(f(x))^{n+1}$ now diffrentiate both sides ; [Ans: $f(x) = x^{1/n}$]



12. Show that the curve passing through (1, 2) for which the segment of the tangent between P and T(on x-axis) is bisected at its point of intersection with the y-axis is a parabola.

Ans: $y^2 = 4x$

Sol



Equation of tangent at P(x, y)

$Y - y = \frac{dy}{dx}(X - x)$

For x-axis, Put $Y = 0 \Rightarrow X = x - \frac{y}{m}$

$\Rightarrow \frac{x - \frac{y}{m} + x}{2} = 0 \Rightarrow 2x = \frac{y}{m}$

$\Rightarrow \frac{dy}{dx} = \frac{y}{2x}$

$\Rightarrow \ell n|y| = \frac{1}{2}\ell n|x| + \ell nC$

Passes through (1, 2)

$\Rightarrow \ell n2 = \ell nC$

$\Rightarrow 2\ell n|y| = \ell n|x| + 2\ell n2$

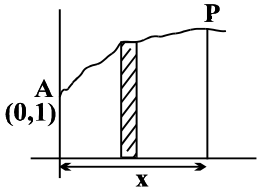
$\Rightarrow y^2 = 4x$

13. Find the curve for which area bounded by the curve, the co-ordinate axes and a variable ordinate is equal to the length of the corresponding arc. Given that the curve passes through (0, 1). **[O]**

Ans: $y = \dfrac{e^x + e^{-x}}{2}$ & $y = 1$]

Sol $\therefore \int_0^x y\,dx = \int_A^P dl = \int \sqrt{(dx)^2 + (dy)^2} = \int_0^x \sqrt{1 + \left(\dfrac{dy}{dx}\right)^2}\,dx$

Now diffrentiate both sides.



14. The rate of growth of bacteria in a culture is proportional to the number of bacteria present and the bacteria count is 1000 at initial time t = 0. the number of bacteria is increased by 20% in 2 hours. If the population of bacteria is 2000 after $\dfrac{k}{\log_e\left(\dfrac{6}{5}\right)}$ hours, then $\left(\dfrac{k}{\log_e 2}\right)^2$ is equal to : **[JEE(Main)Feb.2021]**

Ans. 4

Sol. Let number of bacteriat at any time t is N(t)

$\dfrac{dN}{dt} = \lambda N$ Where λ is a constant.

$\Rightarrow \int \dfrac{dN}{N} = \int \lambda dt$

$\Rightarrow \log_e N = \lambda t + C$

given at t = 0 ; N = 1000

$\Rightarrow C = \log_e 1000$

Also in t = 2 ; N = 1200

So $\log_e 1200 = \lambda \times 2 + \log_e 1000$

$\Rightarrow \lambda = \dfrac{1}{2} \log_e \dfrac{6}{5}$

If N = 2000

$\Rightarrow \log_e 2000 = \dfrac{1}{2}\log_e \dfrac{6}{5}(t) + \log_e 1000$

$\Rightarrow t = \dfrac{2\log_e 2}{\log_e\left(\dfrac{6}{5}\right)} = \dfrac{k}{\log_e\left(\dfrac{6}{5}\right)} \Rightarrow k = \log_e 4$

$$\left(\frac{k}{\log_e 2}\right)^2 = \left(\frac{\log_e 4}{\log_e 2}\right)^2 = (2)^2 = 4$$

15. Let $f(x) = \int_0^x e^t f(t)dt + e^x$ be a differentiable function for all $x \in R$. Then $f(x)$ equals :

[JEE(Main)Feb.2021]

Ans. $2e^{(e^x-1)} - 1$

Sol. $f(x) = \int_0^x e^t f(t)\,dt + e^x$ .....(i)

Differentiate (i) w.r.t. 'x'

$f'(x) = e^x f(x) + e^x$

$\Rightarrow f'(x) = e^x (f(x) + 1)$

$$\Rightarrow \frac{f'(x)}{f(x)+1} = e^x$$

Integrating both sides

$$\int \frac{f'(x)}{f(x)+1}dx = \int e^x dx$$

$\Rightarrow \ln (f(x) + 1) = e^x + c$

$f(x) + 1 = e^{(e^x+c)}$

$f(x) = e^{e^x} \cdot e^c - 1$ .....(ii)

Put $x = 0$ in (i)

$f(0) = 0 + e^\circ = 1$

Put $x = 0$ in (ii)

$f(0) = e.e^c - 1$

$\Rightarrow 1 = e.e^c - 1$

$\Rightarrow e.e^c = 2$

$$\because e^c = \frac{2}{e}$$

$$f(x) = e^{e^x} \cdot \frac{2}{e} - 1$$

$$= 2\left(e^{(e^x-1)}\right) - 1$$

16. Let $y = y_1(x)$ and $y = y_2(x)$ be the solution curves of the differential equation $\frac{dy}{dx} = y + 7$ with initial conditions $y_1(0) = 0$ and $y_2(0) = 1$ respectively. Then the curves $y = y_1(x)$ and $y = y_2(x)$ intersect at.

Ans. no point

Sol. $\frac{dy}{dx} = y + 7 \Rightarrow \frac{dy}{dx} - y = 7$

I.F. $= e^{-x}$

$ye^{-x} = \int 7e^{-x}dx$

$\Rightarrow ye^{-x} = -7e^{-x} + c$

$\Rightarrow y = -7 + ce^{x}$

$-7 + 7e^{x} = -7 + 8e^{x} \Rightarrow e^{x} = 0$

No solution.

2.2 VARIABLE SEPARABLE - TYPE 2

Consider the differential equation $\frac{dy}{dx} = f(ax + by + c)$ . . . (i)

where f(ax + by + c) is some function of 'ax + by + c'.

Let t = ax + by + c.

$\therefore \quad \frac{dt}{dx} = a + b\frac{dy}{dx}$ or $\frac{dy}{dx} = \frac{\frac{dt}{dx} - a}{b}$.

$\therefore \quad$ (i) $\frac{\frac{dt}{dx} - a}{b} = f(t) \Rightarrow \frac{dt}{dx} = bf(t) + a \Rightarrow \frac{dt}{bf(t) + a} = dx$ . . . (ii)

In the differential equation (ii), the variables x and t are separated. Integrating (ii), we get

$$\int \frac{dt}{bf(t) + a} = \int dx + C$$

$$\Rightarrow \int \frac{dt}{bf(t) + a} = x + C \text{ where } t = ax + by + c$$

1. $(x + y)^2 \frac{dy}{dx} = a^2$

Ans. $y = a\tan^{-1}\left(\frac{x + y}{a}\right) + C$

Sol (start with $x + y = t$)

$$(x+y)^2\frac{dy}{dx}=a^2$$

Put $x+y=u \Rightarrow 1+\frac{dy}{dx}=\frac{du}{dx}$

$$\Rightarrow u^2\left(\frac{du}{dx}-1\right)=a^2$$

$$\Rightarrow u^2\frac{du}{dx}=a^2+u^2$$

$$\Rightarrow \frac{u^2}{a^2+u^2}du=dx$$

$$\Rightarrow u-a^2\cdot\frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right)=x+c$$

$$\Rightarrow x+y-a\tan^{-1}\left(\frac{u}{a}\right)=x+c$$

$$\Rightarrow y=c+a\tan^{-1}\left(\frac{x+y}{a}\right)$$

2. $\sin^{-1}\left(\frac{dy}{dx}\right)=x+y$

Ans. $\frac{-2}{1+\tan\left(\frac{x+y}{2}\right)}=x+C$

Sol (start with $x+y=t$)

$$\sin^{-1}\left(\frac{dy}{dx}\right)=x+y$$

Put $x+y=u$

$$\Rightarrow \frac{du}{dx}-1=\sin u$$

$$\Rightarrow \frac{du}{1+\sin u}=dx$$

$$\Rightarrow \frac{\left(1+\tan^2\frac{u}{2}\right)du}{\left(1+\tan\frac{u}{2}\right)^2}=dx$$

$$\Rightarrow \frac{-2}{1+\tan\frac{u}{2}}=x+c \Rightarrow \frac{-2}{1+\tan\left(\frac{x+y}{2}\right)}=x+C$$

3. $\sqrt{1+x+y}\,\dfrac{dy}{dx}=x+y-1$

Ans. $2t-\ln\left|t^2+t-2\right|+\dfrac{5}{3}\ln\left|\dfrac{t-1}{t+2}\right|=x+C \quad (t=\sqrt{x+y+1})$

Sol (put $x+y+1=t^2$)

$\sqrt{1+x+y}\,\dfrac{dy}{dx}=x+y-1$

Let $x+y+1=u^2 \Rightarrow 1+\dfrac{dy}{dx}=2u\dfrac{du}{dx}$

$\Rightarrow u\left(2u\dfrac{du}{dx}-1\right)=u^2-2$

$\Rightarrow 2u^2\dfrac{du}{dx}=u^2+u-2$

$\Rightarrow \dfrac{2u^2}{u^2+u-2}du=dx$

$\Rightarrow \dfrac{2\left(u^2+u-2-u+2\right)}{u^2+u-2}du=dx$

$\Rightarrow 2-\dfrac{2(u-2)}{u^2+u-2}du=dx$

$\Rightarrow 2-\dfrac{(2u+1-5)}{u^2+u-2}du=dx$

$\Rightarrow 2-\dfrac{(2u+1)}{u^2+u-2}+\dfrac{5}{\left(u+\frac{1}{2}\right)^2-\left(\frac{3}{2}\right)^2}du=dx$

$\Rightarrow 2u-\ell n\left|u^2+u-2\right|+\dfrac{5}{2\cdot\frac{3}{2}}\ell n\left|\dfrac{u+\frac{1}{2}-\frac{3}{2}}{u+\frac{1}{2}+\frac{3}{2}}\right|=x+C$

$\Rightarrow 2u-\ell n\left|u^2+u-2\right|+\dfrac{5}{3}\ell n\left|\dfrac{u-1}{u+1}\right|=x+c \qquad x=\sqrt{x+y}+1$

4. $\dfrac{dy}{dx}=\cos(10x+8y)\ ;\ y(0)=0$

Ans. $4y=\tan^{-1}(3\tan 3x)-5x$

Sol. $\dfrac{dy}{dx}=\cos(10x+8y)$

Put $10x + 8y = u \Rightarrow 10 + 8\dfrac{dy}{dx} = \dfrac{du}{dx}$

$$\Rightarrow \frac{\frac{du}{dx} - 10}{8} = \cos(u)$$

$$\Rightarrow \frac{du}{8\cos u + 10} = dx$$

$$\Rightarrow \frac{\sec^2\frac{u}{2}\,du}{2\tan^2\frac{u}{2} + 18} = dx$$

Put $\tan\dfrac{u}{2} = z \Rightarrow \sec^2\dfrac{u}{2}\cdot\dfrac{1}{2}du = dz$

$$\Rightarrow \int \frac{dz}{z^2 + 9} = dx$$

$$\Rightarrow \frac{1}{3}\tan^{-1}\left(\frac{\tan\frac{u}{2}}{3}\right) = x + c$$

$$\Rightarrow \tan^{-1}\left(\frac{\tan\frac{u}{2}}{3}\right) = 3x + 3C$$

When $x = 0 \Rightarrow y = 0 \Rightarrow u = 0 \Rightarrow c = 0$

$$\Rightarrow \frac{\tan\frac{u}{2}}{3} = \tan(3x)$$

$\Rightarrow u = 2\tan^{-1}(3\tan 3x)$

$\Rightarrow 5x + 4y = \tan^{-1}(3\tan 3x)$

5. $\dfrac{dy}{dx} = \dfrac{2x + 3y - 1}{4x + 6y - 5}$

Ans. $\dfrac{2}{7}(2x + 3y - 1) - \dfrac{9}{49}\ln\left|7(2x + 3y - 1) - 6\right| = x + C$

Sol. $\dfrac{dy}{dx} = \dfrac{2x + 3y - 1}{4x + 6y - 5}$

Put $2x + 3y - 1 = u \Rightarrow 2 + 3\dfrac{dy}{dx} = \dfrac{du}{dx}$

$$\Rightarrow \frac{\frac{du}{dx}-2}{3}=\frac{u}{2u-3}$$

$$\Rightarrow \frac{du}{dx}=2+\frac{3u}{2u-3}=\frac{7u-6}{2u-3}$$

$$\Rightarrow \frac{2u-3}{7u-6}\cdot du=dx$$

$$\Rightarrow \frac{\frac{2}{7}(7u-6)}{7u-6}\cdot\frac{-\frac{9}{7}}{7u-6}du=dx$$

$$\Rightarrow \frac{2}{7}u-\frac{9}{49}\ell n|7u-6|=x+C$$

Here $u = 2x + 3y - 1$

2.3 VARIABLE SEPARABLE - TYPE 3

$$\frac{dy}{dx}=\frac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}$$ **If $a_2 + b_1 = 0$**

1. $$\frac{dy}{dx}=\frac{4x-3y}{3x-2y}$$

Ans. $3xy - y^2 - 2x^2 = c$

Sol. $3xdy - 2ydy = 4xdx - 3ydx$

or $3(xdy + ydx) - 2y\,dy - 4xdx = 0$

or $3\,d(xy) - 2ydy - 4xdx = 0$; Integration gives $3xy - y^2 - 2x^2 = c$

2. $$\frac{dy}{dx}=\frac{x-2y+5}{2x+3y-1}$$

Ans. $$2xy=y-\frac{3y^2}{2}+\frac{x^2}{2}+5x+c$$

Sol. $$\frac{dy}{dx}=\frac{x-2y+5}{2x+3y-1}$$

$\Rightarrow 2xdy + 3ydy - dy = xdx - 2ydx + 5dx$

$\Rightarrow 2(xdy + ydx) + 3ydy - dy - xdx - 5dx = 0$

$$\Rightarrow 2xy+\frac{3y^2}{2}-y-\frac{x^2}{2}-5x+c=0$$

2.4 VARIABLE SEPARABLE - TYPE 4

Polar Substitutions

Sometimes transformation to the polar co-ordinates facilitates separation of variables. In this connection it is convenient to remember the following differentials:

(a) If $x = r\cos\theta$; $y = r\sin\theta$ then,

(i) $x\,dx + y\,dy = r\,dr$ (ii) $x\,dy - y\,dx = r^2 d\theta$

(b) If $x = r\sec\theta$ & $y = r\tan\theta$ then

(i) $x\,dx - y\,dy = r\,dr$ (ii) $x\,dy - y\,dx = r^2\sec\theta\,d\theta$.

1. $\dfrac{xdx - ydy}{xdy - ydx} = \sqrt{\dfrac{1 + x^2 - y^2}{x^2 - y^2}}$

Ans. $\sqrt{x^2 - y^2} + \sqrt{1 + x^2 - y^2} = C\left(\dfrac{x + y}{\sqrt{x^2 - y^2}}\right)$

Sol. $\dfrac{xdx - ydy}{xdy - ydx} = \sqrt{\dfrac{1 + x^2 - y^2}{x^2 - y^2}}$

Put $x = r\sec\theta$, $y = r\tan\theta$

$$\Rightarrow \frac{rdr}{r^2\sec\theta d\theta} = \frac{\sqrt{1 + r^2}}{r}$$

$$\Rightarrow \frac{dr}{\sqrt{1 + r^2}} = \sec\theta d\theta \Rightarrow \ell n\left|r + \sqrt{1 + r^2}\right| = \ell n\left|\sec\theta + \tan\theta\right| + \text{ℷ}nC$$

$$\Rightarrow \sqrt{x^2 - y^2} + \sqrt{1 + x^2 - y^2} = \left(\frac{x + y}{\sqrt{x^2 - y^2}}\right)$$

2. $\dfrac{x + y\dfrac{dy}{dx}}{x\dfrac{dy}{dx} - y} = \sqrt{\dfrac{1 - x^2 - y^2}{x^2 + y^2}}$

Ans. $\sin^{-1}\left(\sqrt{x^2 + y^2}\right) = \tan^{-1}\left(\dfrac{y}{x}\right) + C$

Sol. $\dfrac{x + y\dfrac{dy}{dx}}{x\dfrac{dy}{dx} - y} = \sqrt{\dfrac{1 - x^2 - y^2}{x^2 + y^2}}$

Put $x = r\cos\theta$, $y = r\sin\theta$

$$\Rightarrow \frac{rdr}{r^2 d\theta} = \frac{\sqrt{1 - r^2}}{r} \Rightarrow \frac{dr}{\sqrt{1 - r^2}} = d\theta$$

$$\Rightarrow \sin^{-1}(r) = \theta + C$$

$$\Rightarrow \sqrt{x^2 + y^2} = \sin\left(\theta + C\right)$$

where $\theta = \tan^{-1}\left(\dfrac{y}{x}\right)$

3. $xdx + ydy = x(xdy - ydx)$

Ans. $(y+1)^2 = C^2(x^2+y^2)$

Sol. $x = r\cos\theta$; $y = r\sin\theta$; $\tan\theta = \frac{y}{x}$

$\Rightarrow$ $x\,dx + y\,dy = r\,dr$; $xdy - ydy = r^2\,d\theta$

$\therefore$ $rdr = r\cos\theta\, r^2\, d\theta$

$$\int \frac{dr}{r^2} = \int \cos\theta\, d\theta$$

$\Rightarrow$ $-\frac{1}{r} = \sin\theta + C$ or $\frac{1}{\sqrt{x^2+y^2}} + \frac{y}{\sqrt{x^2+y^2}} = C$

$y + 1 = C\sqrt{x^2+y^2}$ or $(y+1)^2 = C^2(x^2+y^2)$]

4. $\frac{xdx + ydy}{\sqrt{x^2+y^2}} = \frac{ydx - xdy}{x^2}$

Ans. $\sqrt{x^2+y^2} + \frac{y}{x} = C$

Sol. Put $x = r\cos\theta$; $y = r\sin\theta$

$\Rightarrow$ $x^2 + y^2 = r^2 \Rightarrow xdx + ydy = rdr$

& $\frac{y}{x} = \tan\theta$ $\Rightarrow$ $\frac{xdy - ydx}{x^2} = \sec^2\theta\, d\theta$

$$\int \frac{rdr}{r} = -\int \sec^2\theta\, d\theta$$

$r = -\tan\theta + C$

$\sqrt{x^2+y^2} + \frac{y}{x} = C$ Ans.

3.1 HOMOGENEOUS DIFFERENTIAL EQUATION

A differential equation of the form $\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)}$

where $f(x,y)$ and $g(x,y)$ are homogeneous functions of x,y and of the same degree, is said to be homogeneous.

This equation may also be reduced to the form $\frac{dy}{dx} = P\left(\frac{y}{x}\right)$

put $y = vx$

$\frac{dy}{dx} = v + x\frac{dv}{dx}$

$v + x\frac{dv}{dx} = P(v)$

$$\frac{dv}{p(v)-v}=\frac{dx}{x}.$$

Questions on Homogeneous Differential Equations:

1. Show that the differential equation $(x-y)\frac{dy}{dx}=x+2y$ is homogeneous and solve it.

Ans. $ln\left(y^2+xy+x^2\right)=2\sqrt{3}\tan^{-1}\left(\frac{2y+x}{\sqrt{3}\,x}\right)+c$

Sol. $(x-y)\frac{dy}{dx}=x+2y$

$$\Rightarrow \frac{dy}{dx}=\frac{x+2y}{x-y}$$

Put $y=ux$

$$\Rightarrow \frac{du}{\frac{1+2u}{1-u}-u}=\frac{dx}{x}\Rightarrow \frac{1-u}{u^2+u+1}du=\frac{dx}{x}$$

$$\Rightarrow \frac{u-1}{u^2+u+1}du=-\frac{dx}{x}$$

$$\Rightarrow \frac{2u+1}{u^2+u+1}-\frac{3}{u^2+u+1}du=-2\frac{dx}{x}$$

$$\Rightarrow \ell n\left|u^2+u+1\right|-\frac{3}{\left(\frac{\sqrt{3}}{2}\right)}\tan^{-1}\left(\frac{u+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)=-2\ell nx+C$$

$$\Rightarrow \ell n\left|\frac{y^2}{x^2}+\frac{y}{x}+1\right|+\ell nx^2=c+2\sqrt{3}\tan^{-1}\left(\frac{2y+x}{\sqrt{3}x}\right)$$

2. Show that the differential equation $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx}=y\cos\left(\frac{y}{x}\right)+x$ is homogeneous and solve it.

Ans. $\sin\left(\frac{y}{x}\right)=ln\,x+C$

Sol. $$\frac{dy}{dx}=\frac{y\cos\left(\frac{y}{x}\right)+x}{x\cos\left(\frac{y}{x}\right)}$$

Put $y=ux$

$$\Rightarrow \frac{du}{\frac{u\cos u+1}{\cos u}-u}=\frac{dx}{x}$$

$$\Rightarrow \cos u\,du=\frac{dx}{x}$$

$$\Rightarrow \sin u=\ell n|x|+\ell nC$$

3. $x^2\frac{dy}{dx}=x^2-2y^2+xy$

Ans. $\frac{1}{2\sqrt{2}}\,ln\left|\frac{\sqrt{2}y+x}{\sqrt{2}y-x}\right|=ln\,x+C$

Sol. $\frac{dy}{dx}=\frac{x^2-2y^2+xy}{x^2}$

Put $y=ux$

$$\Rightarrow \frac{du}{\left(1-2u^2+u\right)-u}=\frac{dx}{x}$$

$$\Rightarrow -\frac{1}{2}\frac{du}{u^2-\left(\frac{1}{\sqrt{2}}\right)^2}=\frac{dx}{x}$$

$$\Rightarrow -\frac{1}{2\cdot\left(\frac{2}{\sqrt{2}}\right)}\ell n\left|\frac{u-\frac{1}{\sqrt{2}}}{u+\frac{1}{\sqrt{2}}}\right|=\ell nx+\ell nC$$

$$\Rightarrow \frac{1}{2\sqrt{2}}\ell n\left|\frac{\sqrt{2}y+x}{\sqrt{2}y-x}\right|=\ell nx+\ell nC$$

4. $\frac{dy}{dx}=\frac{y^2-2xy-x^2}{y^2+2xy-x^2}$ given $y|_{x=1}=-1$

Ans. $x+y=0$

Sol start with $y=vx$ to get $\int\frac{v^2+2v-1}{(1+v)(1+v^2)}\,dv$

$$=\int\frac{2v(v+1)-(1+v^2)}{(1+v)(1+v^2)}\,dv=-\int\frac{dx}{x}$$

Since $v+1$ is cancelled hence another solution is $y=-x$ when $v=-1$

5. $\left(x\frac{dy}{dx} - y\right)\tan^{-1}\frac{y}{x} = x$ given $y|_{x=1} = 0$

Ans. $\sqrt{x^2+y^2} = e^{\frac{y}{x}\tan^{-1}\frac{y}{x}}$

Sol. $x\left(\frac{dy}{dx}-\frac{y}{x}\right).\tan^{-1}\frac{y}{x} = x$ or $\left(\frac{dy}{dx} - \frac{y}{x}\right)\tan^{-1}\frac{y}{x} = 1$

put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ or $\left(v + x\frac{dv}{dx} - v\right)\tan^{-1} v = 1$

$x\frac{dv}{dx}\tan^{-1} v = 1$ or $\int \tan^{-1} v \; dv = \int \frac{dx}{x}$

$v\tan^{-1}v - \int \frac{v}{1+v^2}\,dv = \ell nx + C$

$v.\tan^{-1}v - \frac{1}{2}\ell n(1+v^2) = \ell nx + C$

where $x = 1$; $y = 0 \Rightarrow v = 0 \Rightarrow C = 0$

$\frac{y}{x}\tan^{-1}\frac{y}{x} = \ell n\sqrt{1+\frac{y^2}{x^2}} + \ell nx = \ell n\, x\sqrt{1+\frac{y^2}{x^2}} = \ell n\sqrt{x^2+y^2}$

$\therefore \sqrt{x^2+y^2} = e^{\frac{y}{x}\tan^{-1}\frac{y}{x}}$]

6. A solution of the equation $x\frac{dy}{dx} = y(\log y - \log x + 1)$ is

Ans. $y = xe^{cx}$

Sol. Putting $y = vx$ in the given equation, we have

$v + x\frac{dv}{dx} = v(\log v + 1)$ $\Rightarrow$ $x\frac{dv}{dx} = v\log v$

$\Rightarrow$ $\frac{dv}{v\log v} = \frac{dx}{x}$ $\Rightarrow$ $\log|\log v| = \log|x| + \text{constant}.$

$\Rightarrow$ $\log v = \pm Ax = cx$ $\Rightarrow$ $y = xe^{cx}$.

7. $y\left(\frac{dy}{dx}\right)^2 + 2x\frac{dy}{dx} - y = 0;\; y|_{x=0} = \sqrt{5}$

Ans. $y = \sqrt{5 \pm 2\sqrt{5}\,x}$

Sol. Sometimes a Homogeneous differential equation is given in the form of a quadratic equation in $\frac{dy}{dx}$ (i.e. degree 2,)

$$\frac{dy}{dx} = \frac{-2x \pm \sqrt{4x^2+4y^2}}{2y} = \frac{dy}{dx} = \frac{-x \pm \sqrt{x^2+y^2}}{y}$$ which is homogeneous

put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$$v + x\frac{dv}{dx} = \frac{-x \pm x\sqrt{1+v^2}}{vx} = \frac{-1 \pm \sqrt{1+v^2}}{v}$$

$$\therefore x\frac{dv}{dx} = \frac{-1 \pm \sqrt{1+v^2}}{v} - v = \frac{-1 \pm \sqrt{1+v^2} - v^2}{v}$$

Taking +ve sign

$$\int \frac{v\,dv}{\sqrt{v^2+1} - (1+v^2)} = \int \frac{dx}{x} \quad \text{or} \quad \int \frac{v\,dv}{\sqrt{v^2+1}\left(\sqrt{1+v^2}-1\right)} = -\int \frac{dx}{x}$$

$$\sqrt{1+v^2} - 1 = t \quad \text{or} \quad \frac{v}{\sqrt{1+v^2}}dv = dt$$

$$\int \frac{dt}{t} = -\int \frac{dx}{x}$$

$ln\, tx = C$

$$ln\left[\left(\sqrt{1+\frac{y^2}{x^2}}-1\right)x\right] = C \quad \text{or} \quad ln\left[\sqrt{x^2+y^2}-x\right] = C$$

put $x = 0$; $y = \sqrt{5}$

$$ln\left[\sqrt{5}-0\right] = C \Rightarrow C = ln\sqrt{5}$$

$$\therefore \sqrt{x^2+y^2} = \sqrt{5} + x$$

$$x^2 + y^2 = 5 + x^2 + 2\sqrt{5}\,x$$

$$\therefore y^2 = 5 + 2\sqrt{5}\,x \quad \Big] \; y = \sqrt{5 \pm 2\sqrt{5}\,x}$$

|||ly with –ve sign , $y^2 = 5 - 2\sqrt{5}\,x$]

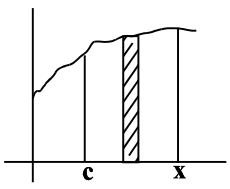
8. The area of the figure bounded by a curve, the x-axis and two ordinates, one of which is constant, the other is variable is equal to the ratio of the cube of the variable ordinate to the variable abscissa. Find the equation of curve if it passes through (1, 0).

Ans. $(2y^2 - x^2)^3 = x^2$

Sol. $\int_c^x y\,dx = \frac{y^3}{x}$

$$y = \frac{x.3y^2\,y_1 - y^3}{x^2}$$

$x^2 = 3xyy_1 - y^2$ $(y \neq 0)$



$\therefore \frac{dy}{dx} = \frac{x^2+y^2}{3xy}$ which is Homogeneous.

9. Find the equation of the curve intersecting with the x-axis at the point x = 1 and possesses the property that the length of the subnormal at any point of the curve is equal to the arithmetic mean of the co-ordinates of this point

Ans: $(y-x)^2 (x+2y) = 1$

Sol. $S.N. = \left| y \frac{dy}{dx} \right| = \frac{x+y}{2} \Rightarrow \frac{dy}{dx} = \frac{x+y}{2y}$

10. Find the curve such that the ordinate of any of its points is the proportional mean between the abscissa and the sum of the abscissa and subnormal at the point.

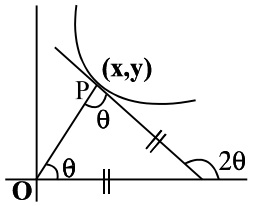
Ans: $y^2 = \frac{x^4+c}{2x^2}$ or $y^2 + 2x^2 \ln x = cx^2$

Sol. $y^2 = x(x + |my|) \Rightarrow y^2 - x^2 = x|my| \Rightarrow \frac{dy}{dx} = \pm \frac{y^2-x^2}{xy}$

11. Find the curve such that the angle, formed with the x-axis by the tangent to the curve at any of its points, is twice the angle formed by the polar radius of the point of tangency with the x-axis. Interpret the curve.

Ans : $x^2 + y^2 - 2cy = 0$]

Sol. Given $\tan 2\theta = \frac{dy}{dx}$ where $\tan\theta = \frac{y}{x}$



$$\frac{2\tan\theta}{1-\tan^2\theta} = \frac{dy}{dx}$$

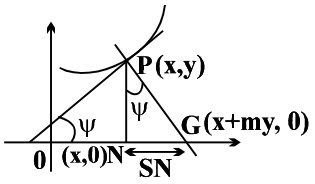
or $$\frac{dy}{dx} = \frac{2\frac{y}{x}}{1-\frac{y^2}{x^2}} = \frac{2xy}{x^2-y^2} \qquad \left(\int \frac{1-v^2}{v(1+v^2)} dv = \int \frac{1+v^2-2v^2}{v(1+v^2)} dv \right)$$

Note that the curve is a circle with centre on y-axis and touching the x-axis at the origin.

12. Find the curve such. that the ratio of the subnormal at any point to the sum of its abscissa and ordinates is equal to the ratio of the ordinate of this point to its abscissa. If the curve passes through (1, 0) find all possible equations in the form y = f(x). **[O]**

Ans. $y = x \ln|x|$; $x^2 + 2xy = 1$

Sol. Normal : $Y - y = -\frac{1}{m}(X - x)$



$$\frac{|my|}{x+y} = \frac{y}{x}$$

$$\frac{dy}{dx} = \pm\frac{x+y}{x}$$

with +ve sign solution is $y = x \ln|x|$

with –ve sign solution is $x^2 + 2xy = 1$

13. Find the curve for which the sum of the normal and subnormal is proportional to the abscissa, the proportionality constant being 1. **[O]**

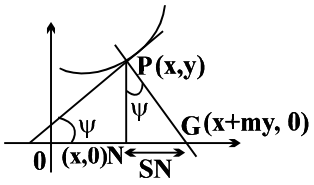
Ans. $-\frac{1}{3}\ln|1-3v^2| = \ln x + C$ or $\ln|1+v^2| = -\ln x + C$

$\left|x^2 - 3y^2\right| x = c$ or $x^2 + y^2 = cx$

Sol. Refer to the figure of

$$\sqrt{m^2 y^2 + y^2} + |my| = kx$$

$$|y|\sqrt{1+m^2} = kx - |my|$$



squaring $y^2 (1+m^2) = k^2x^2 + m^2 y^2 - 2kx\,|my|$

$\therefore \quad 2kx\,|my| = k^2 x^2 - y^2$

$$\frac{dy}{dx} = \pm\frac{k^2x^2 - y^2}{2kxy} \quad \text{Put } k = 1$$

3.2 EQUATIONS REDUCIBLE TO HOMOGENEOUS DIFFERENTIAL EQUATION

$$\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2} = \frac{L_1}{L_2}$$

Case 1 **$a_2 + b_1 = 0$ Variable separable T - 3**

Case 2 L_1 is parallel to L_2 Variable Separable T - 2

Case 3 L_1 and L_2 are intersecting lines

$$\frac{dy}{dx} = \frac{x + 2y - 4}{2x + y - 5}$$

get POI of L_1 and L_2

$$\left.\begin{aligned} x + 2y - 4 = 0 \\ 2x + y - 5 = 0 \end{aligned}\right] \quad x = 2,\ y = 1$$

$x - 2 = X \qquad y - 1 = Y$

$$\frac{dy}{dx} = \frac{dY}{dX}$$

$$\frac{dY}{dX} = \frac{X + 2Y}{2X + Y}$$

$Y = vX$

Extension :

$$\frac{dy}{dx} = \left(\frac{x + 2y - 4}{2x + y - 5}\right)^2$$

$x - 2 = X, \qquad y - 1 = Y$

$$\frac{dY}{dX} = \left(\frac{X + 2Y}{2X - Y}\right)^2$$

4.1 LINEAR DIFFERENTIAL EQUATION

A differential equation is said to be linear if the dependent variable & all its differential coefficients occur in degree one only and are never multiplied together.

The nth order linear differential equation is of the form ;

$a_0(x)\frac{d^n y}{dx^n} + a_1(x)\frac{d^{n-1}y}{dx^{n-1}} + ... + a_n(x).y = \phi(x)$. Where $a_0(x)$, $a_1(x)$ ..... $a_n(x)$ are called the coefficients of the differential equation .

Note that a linear differential equation is always of the first degree but every differential equation of the first degree need not be linear. e.g. the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 + y^2 = 0$ is not linear, though its degree is 1.

Linear Differential Equations Of First Order

Standard Form $\frac{dy}{dx} + Py = Q$, where P, Q are functions of x alone

How to solve first order linear differential equation

1. Convert the given D.E. in standard form.
2. Compute integrating factor $= e^{\int P dx}$
3. Multiply both sides by IF then LHS becomes derivative of y(IF)

$$e^{\int P dx}\left(\frac{dy}{dx} + Py\right) = Q.e^{\int P dx}$$

Or, $\frac{dy}{dx}\cdot e^{\int P dx} + y.\frac{d}{dx}e^{\int P dx} = Qe^{\int P dx}$ $\left[\text{since } \frac{d}{dx}.\left(e^{\int P dx}\right) = P.e^{\int P dx}\right]$

Or, $\frac{d}{dx}\left(y.e^{\int P dx}\right) = Q.e^{\int P dx}$ or,

4. Integrate both sides.

$$\int \frac{d}{dx}.\left(y.e^{\int P dx}\right)dx = \int Q.e^{\int P dx}\,dx$$

Or, $y.e^{\int P dx} = \int Q.e^{\int P dx} + C$

Questions

1. $\frac{dy}{dx} - y = \cos x.$

Ans. $ye^{-x} = \frac{e^{-x}}{2}(\sin x - \cos x) + C$

Sol. $\frac{dy}{dx} - y = \cos x$

$\Rightarrow$ I.F. $= e^{\int -dx} = e^{-x}$

$$ye^{-x} = \int \cos x e^{-x} dx$$

$$\Rightarrow ye^{-x} = \frac{1}{2}e^{-x}\left(-\cos x + \sin x\right) + C$$

2. $\frac{dy}{dx} + 3y = e^{-2x}$

Ans. $ye^{3x} = e^{x} + C$

Sol. $\frac{dy}{dx} + 3y = e^{-2x}$

$$\text{I.F.} = e^{\int 3dx} = e^{3x}$$

$$\Rightarrow ye^{3x} = \int e^{3x} \cdot e^{-2x} dx$$

$$\Rightarrow ye^{3x} = e^{x} + C$$

3. $\cos^2 x \frac{dy}{dx} + y = \tan x \left(0 \le x < \frac{\pi}{2}\right)$

Ans. $ye^{\tan x} = e^{\tan x}\left(\tan x - 1\right) + C$

Sol. $\cos^2 x \frac{dy}{dx} + y = \tan x$

$$\Rightarrow \frac{dy}{dx} + y\sec^2 x = \tan x \sec^2 x$$

$$\Rightarrow \text{I.F.} = e^{\int \sec^2 x dx} = e^{\tan x}$$

$$\Rightarrow ye^{\tan x} = \int e^{\tan x} \cdot \tan x \sec^2 x dx$$

$$= e^{\tan x}(\tan x - 1) + C$$

4. $x \log x \frac{dy}{dx} + y = \frac{2}{x}\log x$

Ans. $y\ell n\, x = -\frac{2}{x}\left(\ell n\left(x\right) + 1\right) + C$

Sol. $x\ell nx \frac{dy}{dx} + y = \frac{2}{x}\ell nx$

$$\Rightarrow \frac{dy}{dx} + \frac{1}{x\ell nx} y = \frac{2}{x^2}$$

$$\text{I.F.} = e^{\int \frac{1}{x\ell nx} dx} = e^{\ell n(\ell nx)} = \ell nx$$

$$\Rightarrow y\ell nx = \int \ell nx \cdot \frac{2}{x^2} dx$$

$$\Rightarrow y\ell nx = \left(\ell nx\right)\left(-\frac{2}{x}\right) - \int \frac{1}{x} \cdot \left(\frac{-2}{x}\right) dx$$

$$= \frac{-2\ell nx}{x} + 2\left(-\frac{1}{x}\right) + C$$

5. $t(1+t^2)\,dx = (x + xt^2 - t^2)\,dt \;\; ; \; x\big|_{t=1} = -\frac{\pi}{4}$

Ans. $x = -t\tan^{-1}t$

Sol. $\frac{dx}{dt} = \frac{x(1+t^2)-t^2}{t(1+t^2)} = \frac{x}{t} - \frac{t}{1+t^2}$

$\frac{dx}{dt} - \frac{1}{t}x = -\frac{t}{1+t^2}$; I.F. $e^{-\int \frac{1}{t}dt} = e^{-ln\,t} = \frac{1}{t}$

$\frac{1}{t}\frac{dx}{dt} - \frac{1}{t^2}x = -\frac{1}{1+t^2} \Rightarrow \frac{d}{dt}\left(\frac{1}{t}.x\right) = -\frac{1}{1+t^2}$

$\frac{x}{t} = -\int \frac{dt}{1+t^2} = -\tan^{-1}t + c \Rightarrow x = t(c - \tan^{-1}t)$

$t = \phi \; ; x = -\frac{\pi}{4} \Rightarrow c = 0$ hence $x = -t\tan^{-1}t$ Ans

6. $x(x^2+1)\frac{dy}{dx} = y(1-x^2) + x^2\,ln x$

Ans. $y\left(\frac{x^2+1}{x}\right) = x(ln\,x - 1) + C$

Sol. $\frac{dy}{dx} = \frac{y(1-x^2)}{x(1+x^2)} + \frac{x\,ln x}{1+x^2}$

$\frac{dy}{dx} - \frac{(1-x^2)y}{x(1+x^2)} = \frac{x\,ln x}{1+x^2}$

I.F. $e^{\int \frac{x^2-1}{x(x^2+1)}dx} = e^{\int \frac{1-\frac{1}{x^2}}{x+\frac{1}{x}}} = e^{ln\left(x+\frac{1}{x}\right)} = x + \frac{1}{x}$

$\therefore \; y.\left(\frac{x^2+1}{x}\right) = \int \frac{x\,ln\,x}{x^2+1}.\frac{x^2+1}{x}\,dx \Rightarrow \left(\frac{x^2+1}{x}\right)y = \int ln\,x\,dx + c$

7. $x\,ln x\frac{dy}{dx} + y = 2\,ln x$

Ans. $y\,ln\,x = ln^2 x + c$

Sol. $x\,ln x\frac{dy}{dx} + y = 2ln\,x$

$\frac{dy}{dx} + \frac{1}{x\,ln x}y = \frac{2}{x}$ I.F. $e^{\int \frac{1}{x\,ln\,x}dx} = e^{ln(ln\,x)} = ln\,x$

$$\therefore \quad l\text{nx}\,\frac{dy}{dx}+\frac{1}{x}y=\frac{2}{x}.l\text{nx}$$

$$\therefore \quad \frac{d}{dx}(y.l\text{nx})=\frac{2}{x}l\text{nx}$$

$$y\,l\text{nx}=2\int\frac{l\text{nx}}{x}\,dx+c$$

8. $(x^2-1)\sin x\,\frac{dy}{dx}+[\,2x\sin x+(x^2-1)\cos x\,]\,y-(x^2-1)\cos x=0.$

Ans. $y(x^2-1)\sin x=(x^2-3)\sin x+2x\cos x+c$

Sol. $\left(x^2-1\right)\sin x\frac{dy}{dx}+\left(2x\sin x+\left(x^2-1\right)\cos x\right)y=\left(x^2-1\right)\cos x$

$$\Rightarrow\frac{d}{dx}\left(y\left(x^2-1\right)(\sin x)\right)=\left(x^2-1\right)\cos x$$

Integrate w.r.t. x

$\Rightarrow y(x^2-1)(\sin x)=(x^2-1)(\sin x)-(2x)(-\cos x)+2(-\sin x)+C$

$=\sin x(x^2-3)+2x\cos x+C$

9. Find the curve such that the initial ordinate of any tangent is less than the abscissa of the point of tangency by two units.

Ans: $y=cx-x\,l\text{n}\,|x|-2$

Sol. Equation of tangent at P(x, y) as curve is

$Y-y=m(X-x)$

$\Rightarrow$ Initial ordinate $=y-mx$

$\Rightarrow y-mx=x-2$

$$\Rightarrow y-\frac{dy}{dx}x=x-2$$

$$\Rightarrow x\frac{dy}{dx}-y=2-x\Rightarrow\frac{dy}{dx}+\left(-\frac{1}{x}\right)y=\frac{2}{x}-1$$

$$\text{I.F.}=e^{\int-\frac{1}{x}dx}=e^{-\ell nx}=\frac{1}{x}$$

$$\Rightarrow y\cdot\frac{1}{x}=\int\frac{1}{x}\left(\frac{2}{x}-1\right)dx$$

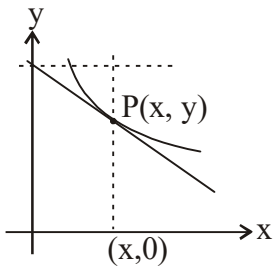
$$\Rightarrow\frac{y}{x}=-\ell n|x|+2\left(-\frac{1}{x}\right)+C$$

$\Rightarrow y=cx-x\,\ell n|x|-2$

10. Find the curve such that the area of the rectangle constructed on the abscissa of any point and the initial ordinate of the tangent at this point is a^2.

Ans: $y=cx\pm\frac{a^2}{2x}$

Sol.



Equation of tangent at P(x, y)

$Y - y = m(X - x)$

Area of rectangle $= |(y - mx)(x)| = a^2$

$$\Rightarrow xy - x^2\frac{dy}{dx} = \pm a^2$$

$$\Rightarrow x^2\frac{dy}{dx} - xy = \pm a^2$$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{x} = \pm\frac{a^2}{x^2}$$

$$\text{I.F.} = e^{\int -\frac{1}{x}dx} = e^{-\ell nx} = -\frac{1}{x}$$

$$\Rightarrow y\left(-\frac{1}{x}\right) = \int \pm\frac{a^2}{x^2}\left(-\frac{1}{x}\right)dx$$

$$\Rightarrow -\frac{y}{x} = \pm a^2\left(\frac{x^{-2}}{-2}\right) + c$$

$$\Rightarrow -y = cx + \frac{a^2}{2x}$$

11. Find the equation of a curve passing through the point (0, 1). If the slope of the tangent to the curve at any point (x, y) is equal to the sum of the x coordinate (abscissa) and the product of the x coordinate and y coordinate (ordinate) of that point.

Ans. $e^{-\frac{x^2}{2}}(y+1) = 2$

Sol. $$\frac{dy}{dx} = x + xy$$

$$\frac{dy}{y+1} = xdx \Rightarrow \ell n(y+1) = \frac{x^2}{2} + C$$

Passes through (0, 1)

$\Rightarrow \ell n2 = C$

$$\Rightarrow \ell n\left(\frac{y+1}{2}\right) = \frac{x^2}{2}$$

$$\Rightarrow \frac{y+1}{2} = 2^{\frac{x^2}{2}}$$

12. $\dfrac{dy}{dx} = \dfrac{1}{x\cos y + \sin 2y}$

Ans. $xe^{-\sin y} = -2e^{-\sin y}(\sin y + 1) + C$

Sol. $\dfrac{dy}{dx} = \dfrac{1}{x\cos y + \sin 2y}$

$$\Rightarrow \frac{dx}{dy} = x\cos y + \sin 2y$$

$$\Rightarrow \frac{dx}{dy} - x\cos y = \sin 2y$$

$$\text{I.F.} = e^{\int -\cos y dy} = e^{-\sin y}$$

$$\Rightarrow xe^{-\sin y} = \int e^{-\sin y} 2\sin y\cos y dy$$

$$\Rightarrow xe^{-\sin y} = 2\int e^{-\sin y}\sin y d(\sin y)$$

$$\Rightarrow xe^{-\sin y} = 2\left(\sin y\left(\frac{e^{-\sin y}}{-1}\right) - (1)\left(\frac{e^{-\sin y}}{1}\right)\right) + C$$

$$\Rightarrow xe^{-\sin y} = 2e^{-\sin y}(-\sin y - 1) + C$$

13. $\dfrac{dy}{dx} = \dfrac{y}{2y\, l\text{n}\, y + y - x}$

Ans $xy = y^2 \ln y + c$

Sol. (linear in $\dfrac{dx}{dy}$/variable separable

or $\dfrac{dx}{dy} = \dfrac{2y\, l\text{n}y + y - x}{y}$

or $\dfrac{dx}{dy} + \dfrac{1}{y}x = (2\, l\text{n}y + 1)$

$xy = y^2 \ln y + c$ Ans.]

14. Let the function $l\text{n}(f(x))$ is defined where $f(x)$ exists for $x \geq 2$ and k is fixed (+)ve real number, prove that if $\dfrac{d}{dx}(x\ f(x)) \leq -k\, f(x)$ then $f(x) \leq A\, x^{-1-k}$ where A is independent of x. **[O]**

Ans.

Sol. $x f'(x) + f(x) \leq -k f(x)$

$x f'(x) + (k+1) f(x) \leq 0$

$x^{k+1} f'(x) + (k+1) x^k f(x) \leq 0$

Note : $x\dfrac{dy}{dx} + (K+1)y \leq 0$

$\dfrac{dy}{dx} + \dfrac{K+1}{x}y \leq 0$

IF $= e^{\int P dx} = e^{\int \frac{K+1}{x} dx} = e^{(K+1)\, l\text{n}\, x} = e^{l\text{n}\, x^{K+1}} = x^{K+1}$

$\frac{d[x^{k+1}\cdot f(x)]}{dx} \le 0$ $\quad\quad$ $\therefore\ x^{K+1}\frac{dy}{dx} + x^{K(K+1)y} \le 0$

Let $F(x) = x^{k+1}\cdot f(x)$

$F(x)$ is decreasing $x \ge 2$

$F(x) \le F(2)$ $\quad$ for all $x \ge 2$

$F(x) \le A$

$x^{k+1}\cdot f(x) \le A$

$f(x) \le A\,x^{-1-k}$]

15. Given a function 'g' which has a derivative $g'(x)$ for every real 'x' and which satisfy $g'(0) = 2$ and $g(x+y) = e^y \cdot g(x) + e^x \cdot g(y)$ for all x & y. Find g(x) and determine the range of the function.

Ans. $g(x) = 2x\,e^x, \left[-\frac{2}{e}, \infty\right)$; 2 sq. units]

Sol. Put $x = 0$; $y = 0$ $\Rightarrow$ $g(0) = 0$

and $g'(0) = \underset{h\to 0}{\text{Limit}}\ \frac{g(h)}{h} = 2$

Now $g'(x) = \underset{h\to 0}{\text{Limit}}\ \frac{g(x+h) - g(x)}{h} = \underset{h\to 0}{\text{Limit}}\ \frac{e^h \cdot g(x) + e^x \cdot g(h) - g(x)}{h}$

$= g(x)\ \underset{h\to 0}{\text{Limit}}\ \left(\frac{e^h - 1}{h}\right) + e^x\ \underset{h\to 0}{\text{Limit}}\ \frac{g(h)}{h}$ $\quad$ or $\quad g'(x) = g(x) + 2e^x$

Let $g(x) = y$ $\quad$ then $\frac{dy}{dx} = y + 2e^x$ $\quad\Rightarrow\quad$ $e^{-x}\frac{dy}{dx} - y\,e^{-x} = 2$

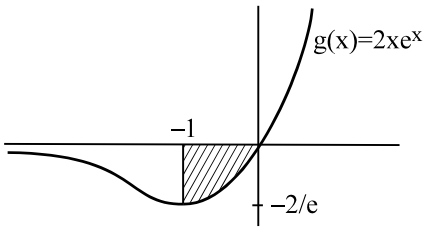
$\Rightarrow\ \frac{d}{dx}(y\,e^{-x}) = 2$ $\quad$ $y\,e^{-x} = 2x + c$; If $\begin{matrix} x = 0 \\ y = 0 \end{matrix}\Big]$ $\Rightarrow$ $c = 0$

Hence $y = 2x\,e^x = g(x)$

Now $\frac{dy}{dx} = 2[e^x + x\,e^x] = 0 \Rightarrow x = -1$

$\Rightarrow$ minima at $x = -1$

$\Rightarrow$ range is $\left[-\frac{2}{e}, \infty\right)$



16. The function y (x) satisfies the equation $y(x) + 2x\int_0^x \frac{y(u)}{1+u^2}du = 3x^2 + 2x + 1$. Show that the substitution $z(x) = \int_0^x \frac{y(u)}{1+u^2}du$ converts the equation into a first order linear differential equation for z(x) solve for z (x) . Hence solve the original equation for y (x). **[O]**

Ans. $z(x) = \frac{x^3 + x^2 + x}{1+x^2}$; $y(x) = \frac{(1+x^2)^2 + 2x}{(1+x^2)}$

Sol. $y(x) + 2x \int_0^x \frac{y(u)}{1+u^2} du = 3x^2 + 2x + 1$ ....(1)

Let $z(x) = \int_0^x \frac{y(u)}{1+u^2} du$

$\therefore \quad z'(x) = \frac{y(x)}{1+x^2} \quad \Rightarrow \quad y(x) = (1 + x^2) z'(x)$ ....(2)

substituting y (x) in equation (1)

$(1 + x^2) z'(x) + 2x \cdot z(x) = 3x^2 + 2x + 1$

Let $z'(x) = \frac{dz}{dx}$

$\frac{dz}{dx} + \frac{2x}{1+x^2} z(x) = \frac{3x^2+2x+1}{1+x^2}$ which is linear DE

I.F. $= e^{\int \frac{2x}{1+x^2} dx} = e^{\ln(1+x^2)} = 1 + x^2$

Hence $z(1 + x^2) = \int (3x^2 + 2x + 1) dx$

$z(1 + x^2) = x^3 + x^2 + x + C$

but $z = 0$ when $x = 0 \Rightarrow C = 0$

Hence $z(x) = \frac{x^3+x^2+x}{1+x^2}$ .....(3)

now $y(x) = (1 + x^2) \cdot z'(x)$ .....(4) (from 2)

also $z(x) = \frac{x(1+x^2)+x^2}{1+x^2} = x + \frac{x^2+1-1}{1+x^2} = x + 1 - \frac{1}{1+x^2}$

$\therefore \quad z'(x) = 1 + \frac{2x}{(1+x^2)^2} = \frac{(1+x^2)+2x}{(1+x^2)^2}$

$\therefore \quad y(x) = \frac{(1+x^2)^2+2x}{(1+x^2)}$ [from eq^n (4)]

17. If a curve y = f (x) passes through the point (1, 2) and satisfies $x\frac{dy}{dx} + y = bx^4$, then for what value of b, $\int_1^2 f(x) dx = \frac{62}{5}$? **[JEE(Main)Feb.2021]**

Ans. 10

Sol. $\frac{xdy}{dx} + y = bx^4$

$$\int d(xy) = \int bx^4 dx$$

$$xy = \frac{bx^5}{5} + c$$

$$y = \frac{bx^4}{5} + \frac{c}{x}$$

Passes though $(1, 2) \Rightarrow \frac{b}{5} + c = 2$ .....(1)

$$\int_1^2 f(x)\,dx = \frac{62}{5}$$

$$\left(\frac{bx^5}{25} + c\ln x\right)_1^2 = \frac{62}{5}$$

$$\frac{31b}{25} + c\ln 2 = \frac{62}{5} \quad .....(2)$$

$\Rightarrow$ solving equation (1) and (2)

$\Rightarrow b = 10$ and $c = 0$

18. If $y = y(x)$ is the solution of the equation $e^{\sin y}\cos y\frac{dy}{dx} + e^{\sin y}\cos x = \cos x, y(0) = 0$; then

$1 + y\left(\frac{\pi}{6}\right) + \frac{\sqrt{3}}{2}y\left(\frac{\pi}{3}\right) + \frac{1}{\sqrt{2}}y\left(\frac{\pi}{4}\right)$ is equal to : **[JEE(Main)Feb.2021]**

Ans. 1

Sol. $e^{\sin y}\cos y\frac{dy}{dx} + e^{\sin y}\cos x = \cos x$

Let $e^{\sin y} = t$

$$\Rightarrow e^{\sin y}\cos y\frac{dy}{dx} = \frac{dt}{dx}$$

$$\Rightarrow \frac{dt}{dx} + t\cos x = \cos x$$

Integrating Factors

I.F. $= e^{\int \cos x dx} = e^{\sin x}$.

$$\Rightarrow \int d(t.e^{\sin x})\,dx = \int e^{\sin x}\cos x\,dx$$

$\Rightarrow t.e^{\sin x} = e^{\sin x} + c$

$\Rightarrow e^{\sin y}.e^{\sin x} = e^{\sin x} + c$

$x = 0;\ y = 0 \Rightarrow c = 0$

$\Rightarrow e^{\sin y}.e^{\sin x} = e^{\sin x}$

$\Rightarrow e^{\sin y} = 1 \Rightarrow \sin y = 0 \Rightarrow y = 0$

$\Rightarrow 1 + y\left(\frac{\pi}{6}\right) + \frac{\sqrt{3}}{2}y\left(\frac{\pi}{3}\right) + \frac{1}{\sqrt{2}}y\left(\frac{\pi}{4}\right) = 1$

19. Let a differentiable function f satisfy $f(x) + \int_3^x \frac{f(t)}{t}dt = \sqrt{x+1}, x \geq 3$. Then $12f(8)$ is equal to :

Ans. 17

Sol. $f'(x) + \frac{f(x)}{x} = \frac{1}{2\sqrt{x+1}}$

$\frac{dy}{dx} + \frac{y}{x} = \frac{1}{2\sqrt{x+1}}$

$I.F. = e^{\int \frac{1}{x}dx} = x$

$y.x = \int x\frac{1}{2\sqrt{x+1}}dx + c$ put $x + 1 = t^2$, $dx = 2tdt$

$y.x = \frac{1}{2}\int \frac{\left(t^2-1\right)2tdt}{t} + c$

$y.x = \frac{1}{2}\left(\frac{2}{3}t^3 - 2t\right) + c$

$y.x = \frac{\left(\sqrt{x+1}\right)^3}{3} - \sqrt{x+1} + c$

now $f(3) = 2$

$6 = \frac{(2)^3}{3} - 2 + c$

$8 - \frac{8}{3} = c$

$\frac{16}{3} = c$

$$\therefore f(x)=\frac{1}{x}\left(\frac{\left(\sqrt{x+1}\right)^3}{3}-\left(\sqrt{x+1}\right)+\frac{16}{3}\right)$$

$$\therefore 12f(8)=\frac{12}{8}\left(\frac{27}{3}-3+\frac{16}{3}\right)$$

$$=\frac{3}{2}\left(6+\frac{16}{3}\right)=17$$

4.2 MIXTURE PROBLEMS :

1. A tank initially contains 100 litres of brine in which 50 gms of salt dissolved. A brine containing 2 gm/litre of salt runs into the tank at the rate of 5 litre/min. The mixture is kept stirring and flows out of the tank at the rate of 4 litres/min then
   (a) At what rate (gms/min) does salt enter the tank at time t.
   (b) What is the volume of the brine in the tank at time t.
   (c) At what rate (gms/min) does salt leave the tank at time t.
   (d) form the DE of the process and solve it to find an expression for the amount of salt present at time t.

Ans. (a) 10 (b) $V(t)=(100+t)$ (c) $\frac{y}{100+t}\cdot 4$ (d) $\frac{dy}{dt}=10-\frac{4y}{100+t}$; $y=2(100+t)-\frac{150}{\left(1+\frac{t}{100}\right)^4}$

Sol.

(a) Rate at which salt is added $=\frac{2\,gm}{litre}\cdot\frac{5\,litres}{min}=10$ gm/min.

(b) Volume V (t) of brine at time $t=100+(5-4)t=(100+t)$ litres

(c) Rate at which salt leaves $=\frac{y(t)}{V(t)}$ (out flow rate) $=\frac{y}{100+t}\cdot 4$

(d) Hence DE is $\frac{dy}{dt}=10-\frac{4y}{100+t}$ which is linear $\frac{dy}{dt}+\frac{4y}{100+t}=10$

I.F. $=e^{+\int\frac{4}{100+t}dt}=e^{+4(ln\,(100+t)}$

$\therefore$ $y(100+t)^4=10\int(100+t)^4\,dt=(100+t)^4+c$

$y(100+t)^4=2(100+t)^5+c$

if $t=0$; $y=50$

$50\cdot 100^4=200\cdot 100^4+c$

$\therefore$ $c=-150\cdot 100^4$

$\therefore$ $y(100+t)^4=2(100+t)^5-150\cdot 100^4$

$y=2(100+t)-\frac{150\cdot 100^4}{(100+t)^4}$; $y=2(100+t)-\frac{150}{\left(1+\frac{t}{100}\right)^4}$]

2. A storage tank contains 2000 litres of gasoline which initially has 100 gms of an additive dissolved in it. Gasoline containing 2 gms of additive per litre is pumped into the tank at a rate of 40 litre/min. The well mixed solution ispumped out at a rate of 45 litre/min.

(a) At what rate (gms/min) does salt enter the tank at time t.
(b) What is the volume of the brine in the tank at time t.
(c) At what rate (gms/min) does salt leave the tank at time t.
(d) form the DE of the process and solve it to find an expression for the amount of salt present at time t.

Ans. (a) 80 (b) $V(t) = (2000 - 5t)$ (c) $\frac{y}{(2000-5t)} \cdot 45$

(d) $\frac{dy}{dt} = 80 - \frac{y}{(2000-5t)} \cdot 45$; $y = 10(400 - t) - 3900\left(1 - \frac{t}{400}\right)^9$

Sol. $\frac{dy}{dt} = 80 - \frac{y}{(2000-5t)} \cdot 45$ $\qquad t = 0$; $y = 100$

$y = 10(400 - t) - 3900\left(1 - \frac{t}{400}\right)^9$ Ans]

3. A tank contains 200 litres of brine in which 20 gms of salt dissolved. Brine containing $\frac{1}{4}$ gm of salt/lit. runs into the tank at the rate of 2 litre/min. The mixture is kept stirring runs out at the same rate.

(a) At what rate (gms/min) does salt enter the tank at time t.
(b) What is the volume of the brine in the tank at time t.
(c) At what rate (gms/min) does salt leave the tank at time t.
(d) form the DE of the process and solve it to find an expression for the amount of salt present at time t.

Ans. (a) 50 (b) $V(t) = 200$ (c) $\frac{y}{200} \cdot 2$ (d) $\frac{dy}{dt} = \frac{1}{2} - \frac{y}{200} \cdot 2$; $y = 50 - 30 \cdot e^{-\frac{t}{100}}$

Sol. $t = 0$; $y = 20$

$\frac{dy}{dt} = \frac{1}{2} - \frac{y}{200} \cdot 2 \quad \Rightarrow \quad \frac{dy}{dx} = \frac{1}{2} - \frac{y}{100} \quad \Rightarrow \quad \frac{50-y}{100}$

Hence

$$\frac{dy}{50-y} = \frac{dt}{100}$$

$$-\ln(50 - y) = \frac{t}{100} + c$$

or $$\ln(50 - y) = -\frac{t}{100} + c$$

$t = 0$; $y = 20$; $c = \ln 30$

Hence $$\ln\frac{50-y}{30} = \frac{t}{100}$$

$$\frac{50-y}{30} = e^{-\frac{t}{100}}$$

$y = 50 - 30 \cdot e^{-\frac{t}{100}}$ Ans.]

4.3 EQUATIONS REDUCIBLE TO LINEAR DIFFERENTIAL EQUATION

4.3.1 BERNOULLI'S EQUATION

$$\frac{dy}{dx} + p(x)\, y = y^n Q(x)$$

$\div$ by y^n

$$\frac{1}{y^n}\frac{dy}{dx} + \frac{p(x)}{y^{n-1}} = Q(x)$$

Put $\frac{1}{y^{n-1}} = z$

$$-\frac{(n-1)}{y^n}\frac{dy}{dx} = \frac{dz}{dx}$$

$$\frac{-1}{(n-1)}\frac{dz}{dx} + P(x)\, z = Q(x)$$

1. $\frac{dy}{dx} = xy + x^3y^2$

Ans. $ze^{x^2/2} = x^2e^{x^2/2} - 2e^{x^2/2} + C$ [where $z = \frac{-1}{y}$]

Sol. $\frac{1}{y^2}\frac{dy}{dx} = \frac{x}{y} + x^3$ or $\frac{1}{y^2}\frac{dy}{dx} - \frac{x}{y} = x^3$

put $-\frac{1}{y} = z \quad \Rightarrow \quad \frac{1}{y^2}\frac{dy}{dx} = \frac{dz}{dx} \Rightarrow \quad \frac{dz}{dx} + z \cdot x = x^3$ which is linear.

2. Find the curve such that the y-intercept cut off by the tangent on any arbitrary point is
   (a) proportional to the square of the ordinate of the point of tangency.
   (b) proportional to the cube of the ordinate of the point of tangency.

Ans. (a) $\frac{a}{x} + \frac{b}{y} = 1$; (b) $\frac{a}{x^2} + \frac{b}{y^2} = 1$]

Sol. Let equation of tangent at P(x, y) on curve is $Y - y = m(X - x)$
Y-intercept : $y - mx$
(a) $y - mx = \lambda y^2$

$$\Rightarrow x\frac{dy}{dx} - y = -\lambda y^2$$

$$\Rightarrow \frac{1}{y^2}\frac{dy}{dx} + \left(-\frac{1}{x}\right)\frac{1}{y} = -\frac{\lambda}{x}$$

Put $z = -\frac{1}{y} \Rightarrow \frac{dz}{dx} = \frac{1}{y^2}\frac{dy}{dx}$

$\Rightarrow \frac{dz}{dx} + \frac{z}{x} = -\frac{\lambda}{x}$

$I.F. = e^{\int \frac{1}{x}dx} = x$

$\Rightarrow zx = \int -\frac{\lambda}{x} x dx$

$\Rightarrow -\frac{x}{y} = -\lambda x + C$

$\Rightarrow -x = \lambda xy + cy$

(b) $y - mx = \lambda y^3$

$\Rightarrow x\frac{dy}{dx} - y = -\lambda y^3$

$\Rightarrow y^{\frac{1}{3}}\frac{dy}{dx} + \frac{1}{x}\cdot\left(-\frac{1}{y^2}\right) = -\frac{\lambda}{x}$

Put $-\frac{1}{y^2} = z \Rightarrow \frac{2}{y^3}\frac{dy}{dx} = \frac{dz}{dx}$

$\Rightarrow \frac{1}{2}\frac{dz}{dx} + \frac{1}{x}z = -\frac{\lambda}{x}$

$\Rightarrow \frac{dz}{dx} + \left(\frac{2}{x}\right)z = -\frac{2\lambda}{x}$

$I.F. = e^{\int \frac{2}{x}dx} = x^2$

$\Rightarrow z\cdot x^2 = \int -\frac{2\lambda}{x}\cdot x^2 dx$

$\Rightarrow -\frac{1}{y^2}x^2 = -\lambda x^2 + C$

$\Rightarrow -x^2 = -\lambda x^2 y^2 + cy^2$

4.3.2 CHANGE OF VARIABLE BY A SUITABLE SUBSTITUTION

1. $y \sin x \frac{dy}{dx} = \cos x (\sin x - y^2)$

Ans. $y^2 \sin^2 x = \frac{2}{3}(\sin x)^3 + C$

Sol. put $y^2 = z \quad \Rightarrow \quad 2y\frac{dy}{dx} = \frac{dz}{dx}$

$$\sin x \frac{dz}{dx} = 2\cos x (\sin x - z)$$

$$\frac{dz}{dx} = 2\cos x - 2\cot x\, z \qquad \text{or} \qquad \frac{dz}{dx} + 2\cot x\, z = 2\cos x \qquad \text{which is linear.}$$

2. $x \frac{dy}{dx} + y \ln y = xye^x$

Ans. $x \ln y = e^x (x - 1) + c$

[Hint: put $\ln y = t$]

3. $\frac{dy}{dx} = e^{x-y}(e^x - e^y)$

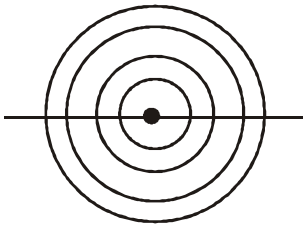
Ans. $e^y = c \cdot \exp(-e^x) + e^x - 1$

[Hint: multiply by e^y & put $e^y = t$]

5.1 ORTHOGONAL TRAJECTORIES

A curve making at each of its points a fixed angle α with the curve of the family passing through that point is called an ***isogonal trajectory*** of that family ; if in particular $\alpha = \pi/2$, then it is called an ***orthogonal trajectory***.

The line passing through origin is orthogonal trajectory for a family of circles centre at origin.



Find the orthogonal trajectories of the following families :

1. $x^2 + y^2 = a^2$

Ans. $y = cx$

Sol step: (i) Form diff. equation representing the given family of curves.

$$x^2 + y^2 = r^2$$

$$2x + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-x}{y}$$

(ii) Replace $\left(\frac{dy}{dx}\right)$ by $\left(\frac{-dx}{dy}\right)$

$$\frac{-dx}{dy} = \frac{-x}{y}$$

(iii) Diff. equation obtained will be the diff. equation of orthogonal trajectory. Integrate it and get equation of orthogonal trajectory.

$$\int \frac{dx}{x} = \int \frac{dy}{y}$$

$l$nx + $l$nc = $l$ny

y = cx

2. $x^2 + y^2 - 2ay = 0$

Ans. $x^2 + y^2 = cx$

Sol. $x^2 + y^2 - 2ay = 0$

differentiating w.r.t. x

$$2x + 2y\frac{dy}{dx} - 2a\frac{dy}{dx} = 0$$

$$\frac{2x + 2y\frac{dy}{dx}}{\frac{dy}{dx}} = 2a$$

$$\Rightarrow x^2 + y^2 - \frac{\left(2x + 2y\frac{dy}{dx}\right)y}{\frac{dy}{dx}} = 0$$

For orthogonal trajectory $\frac{dy}{dx} \to -\frac{dx}{dy}$

$$\Rightarrow \left(x^2 + y^2\right)\left(-\frac{dx}{dy}\right) - y\left(2x - 2y\frac{dx}{dy}\right) = 0$$

$$\Rightarrow -x^2\frac{dx}{dy} - y^2\frac{dx}{dy} - 2xy + 2y^2\frac{dx}{dy} = 0$$

$$\Rightarrow \frac{dx}{dy} = \frac{2xy}{y^2 - x^2} \Rightarrow \frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

Put y = ux

$$\Rightarrow \frac{du}{\frac{u^2 - 1}{2u} - u} = \frac{dx}{x} \qquad \Rightarrow \frac{2u}{u^2 + 1}du = -\frac{dx}{x}$$

$\Rightarrow \ell n|u^2 + 1| = -\ell n|x| + \ell nC$

$\Rightarrow \ell n|(u^2 + 1)x| = \ell nC$

$$\Rightarrow \left(\frac{y^2}{x^2} + 1\right)x = C$$

$\Rightarrow (y^2 + x^2) = cx$

3. $y^2 = 4a\,(x + a)$

Ans. $x + \sqrt{x^2 + y^2} = cy^2$ or $x + \sqrt{x^2 + y^2} = c$

Sol. $2\,y\,y_1 = 4a$ ....(1)

$$y^2 = 2\,y\,y_1\left(x + \frac{yy_1}{2}\right)$$

$$y^2 = 2x\,y\,y_1 + y^2 \cdot y_1{}^2$$

$$y^2\left(\frac{dy}{dx}\right)^2 + 2xy\left(\frac{dy}{dx}\right) - y^2 = 0$$

$$y\left(\frac{dx}{dy}\right)^2 - 2x\left(\frac{dx}{dy}\right) - y = 0$$

$$\frac{dx}{dy} = \frac{2x \pm \sqrt{4x^2 + 4y^2}}{2y} = \frac{x \pm \sqrt{x^2 + y^2}}{y}$$

Let $x = v\,y \qquad \Rightarrow \quad \frac{dx}{dy} = v + y\frac{dv}{dx}$

$$v + y\frac{dv}{dy} = v \pm \sqrt{1 + v^2}$$

$$y = 0$$

$$\int \frac{dv}{\sqrt{1 + v^2}} = \pm \int \frac{dy}{y}$$

$$\ell n\,(v + \sqrt{1 + v^2}\,) = \pm\,\ell n\,(c\,y)$$

$$\frac{x + \sqrt{x^2 + y^2}}{y} = cy$$

$x + \sqrt{x^2 + y^2} = cy^2$ Ans. $\qquad x + \sqrt{x^2 + y^2} = c$ Ans.]

4. $x^2 - \frac{1}{3}y^2 = a^2$

Ans. $xy^3 = c$

Sol. $x^2 - \frac{y^2}{3} = a^2$

differentiating w.r.t. x

$$2x - \frac{2}{3}y\frac{dy}{dx} = 0$$

$$\frac{3x}{y} = \frac{dy}{dx}$$

For orthogonal

$$\frac{3x}{y} = -\frac{dx}{dy} \Rightarrow \frac{3dy}{y} = -\frac{dx}{x}$$

$$\Rightarrow 3\ell n|y| = -\ell n|x| + \ell nC$$

$\Rightarrow y^3x = C$

5. $y = a\,e^{\sigma x}$, where σ is a constant and a is a parameter.

Ans. $\sigma y^2 + 2x = c$

Sol. $y = ae^{bx}$

$$\Rightarrow \frac{dy}{dx} = abe^{bx} \Rightarrow \frac{dy}{dx} = by$$

For orthogonal trajectory

$$-\frac{dx}{dy} = by \Rightarrow -dx = bydy$$

$$\Rightarrow -x = \frac{by^2}{2} + C$$

6. $\cos y = ae^{-x}$ **[O]**

Ans. $\sin y = c\,e^{-x}$

Sol. $\cos y = ae^{-x}$

$$\Rightarrow -\sin y\frac{dy}{dx} = -ae^{-x}$$

$$\Rightarrow \sin y\frac{dy}{dx} = \cos y$$

For orthogonal

$$\Rightarrow -\frac{dx}{dy} = \cot y \Rightarrow -dx = \cot ydy$$

$\Rightarrow -x = \ell n \sin y + \ell nC$

$\Rightarrow C \sin y = e^{-x}$

7. $y^2 = 4(x - a)$ **[O]**

Ans. $y = c\,e^{-x/2}$

Sol. $y^2 = 4(x - a)$

$$\Rightarrow 2y\frac{dy}{dx} = 4 \Rightarrow \frac{dy}{dx} = \frac{2}{y}$$

For orthogonal

$$\Rightarrow -\frac{dx}{dy} = \frac{2}{y} \Rightarrow -dx = 2\frac{dy}{y}$$

$\Rightarrow -x = 2\,\ell ny + \ell nC$

$\Rightarrow -x = \ell ny^2C$

$\Rightarrow y^2 = e^{-x}$

5.2 EXACT DIFFERENTIAL

(i) $xdy + y\,dx = d(xy)$

(ii) $\dfrac{x\,dy - y\,dx}{x^2} = d\left(\dfrac{y}{x}\right)$

(iii) $\dfrac{y\,dx - x\,dy}{y^2} = d\left(\dfrac{x}{y}\right)$

(iv) $\dfrac{x\,dy + y\,dx}{xy} = d(\ln xy)$

(v) $\frac{dx+dy}{x+y} = d(\ln(x+y))$ **(vi)** $\frac{x\,dy-y\,dx}{xy} = d\left(\ln\frac{y}{x}\right)$

(vii) $\frac{y\,dx-x\,dy}{xy} = d\left(\ln\frac{x}{y}\right)$ **(viii)** $\frac{x\,dy-y\,dx}{x^2+y^2} = d\left(\tan^{-1}\frac{y}{x}\right)$

(ix) $\frac{y\,dx-x\,dy}{x^2+y^2} = d\left(\tan^{-1}\frac{x}{y}\right)$ **(x)** $\frac{x\,dx+y\,dy}{x^2+y^2} = d\left[\ln\sqrt{x^2+y^2}\right]$

(xi) $d\left(-\frac{1}{xy}\right) = \frac{x\,dy+y\,dx}{x^2y^2}$ **(xii)** $d\left(\frac{e^x}{y}\right) = \frac{y\,e^x\,dx-e^x\,dy}{y^2}$

(xiii) $d\left(\frac{e^y}{x}\right) = \frac{x\,e^y\,dy-e^y\,dx}{x^2}$

Questions

1. $\left(\frac{1}{y}\sin\frac{x}{y}-\frac{y}{x^2}\cos\frac{y}{x}+1\right)dx+\left(\frac{1}{x}\cos\frac{y}{x}-\frac{x}{y^2}\sin\frac{x}{y}+\frac{1}{y^2}\right)dy=0$

Ans: $\sin\frac{y}{x}-\cos\frac{x}{y}+x-\frac{1}{y}=0$

Sol. Isolating $\sin\frac{x}{y}$ & $\cos\frac{y}{x}$

$$\left(\frac{dx}{y}-\frac{x\,dy}{y^2}\right)\sin\frac{x}{y}+\left(\frac{dy}{x}-\frac{y\,dx}{x^2}\right)\cos\frac{y}{x}+dx+\frac{dy}{y^2}=0$$

$$\frac{ydx-xdy}{y^2}.\sin\frac{x}{y}\quad\frac{xdy-ydx}{x^2}.\cos\frac{y}{x}+dx+\frac{dy}{y^2}=0$$

$$\sin\frac{x}{y}.d\left(\frac{x}{y}\right)+\cos\frac{y}{x}.d\left(\frac{y}{x}\right)+dx+\frac{dy}{y^2}=0$$

$$-\cos\frac{x}{y}+\sin\frac{y}{x}+x-\frac{1}{y}=C \text{ or } \quad \frac{1}{y}+\cos\frac{x}{y}=x+\sin\frac{y}{x}$$

However note that the D.E.

$\left(x\cos\frac{y}{x}+y\sin\frac{y}{x}\right)ydx+\left(x\cos\frac{y}{x}-y\sin\frac{y}{x}\right)xdy=0$ is Homogeneous and can be solved conviniently with $y=vx$

simplifies to $-\int\frac{2\,dx}{x}=-\int\frac{\cos v-v\sin v}{v\cos v}dv$

$\therefore \quad \ln(v\cos v)+\ln x^2=0 \Rightarrow xy\cos\frac{y}{x}=C$]

2. $\left(\frac{1}{x}-\frac{y^2}{(x-y)^2}\right)dx+\left(\frac{x^2}{(x-y)^2}-\frac{1}{y}\right)dy=0$ **[O]**

Ans. $\ell n\left|\frac{x}{y}\right|+\frac{x-y}{xy}=c$

Sol. $\frac{dx}{x}-\frac{dy}{y}+\frac{x^2\,dy-y^2\,dx}{(x-y)^2}=0$ or $\frac{dx}{x}-\frac{dy}{y}+\frac{x^2\,dy-y^2\,dx}{x^2y^2\left(\frac{1}{y}-\frac{1}{x}\right)^2}=0$

or $\frac{dx}{x}-\frac{dy}{y}+\frac{\frac{dy}{y^2}-\frac{dx}{x^2}}{\left(\frac{1}{y}-\frac{1}{x}\right)^2}=0]$

3. $\frac{y+\sin x\cos^2(xy)}{\cos^2(xy)}\,dx+\frac{x\,dy}{\cos^2 xy}+\sin y\,dy=0$

Ans. $\tan(x\,y)=\cos x+\cos y+C]$

Sol. $\frac{x\,dy+y\,dx}{\cos^2 xy}+\sin x\,dx+\sin y\,dy=0$

$\int\frac{d(xy)}{\cos^2 xy}+\int\sin x\,dx+\int\sin y\,dy=0$

$\tan(x\,y)=\cos x+\cos y+C]$

4. $\frac{x+y\frac{dy}{dx}}{y-x\frac{dy}{dx}}=\frac{x\cos^2(x^2+y^2)}{y^3}$

Ans. $\tan(x^2+y^2)=\frac{x^2}{y^2}+c$

Sol. The given differential equation can be written as

$\frac{xdx+ydy}{(ydx-xdy)/y^2}=y^2\frac{x}{y^3}(\cos^2(x^2+y^2)$

$\Rightarrow\quad \sec^2(x^2+y^2)\ \frac{1}{2}d(x^2+y^2)=\frac{x}{y}d\left(\frac{x}{y}\right)\quad\Rightarrow\ \frac{1}{2}\tan(x^2+y^2)=\frac{1}{2}\left(\frac{x}{y}\right)^2+\frac{c}{2}$

$\therefore\quad \tan(x^2+y^2)=\frac{x^2}{y^2}+c$

5. If the curve, $y=y(x)$ represented by the solution of the differential equation $(2xy^2-y)dx+xdy=0$, passes through the intersection of the lines, $2x-3y=1$ and $3x+2y=8$, then $|y(1)|$ is equal to.

[JEE(Main)Feb.2021]

Ans. 1

Sol. $2xy^2\,dx-y\,dx+x\,dy=0$

$$2x\,dx - \frac{1}{y}\,dx + \frac{x}{y^2}\,dy = 0$$

$$2x\,dx = \frac{1}{y}\,dx - \frac{x}{y^2}\,dy$$

$$\int 2x\,dx = \int d\left(\frac{x}{y}\right)$$

$x^2 = \frac{x}{y} + c$ put P.O.I. of $2x - 3y = 1$ and $3x + 2y = 8$

$c = 2$ i.e. $(2, y)$

$y(1) = -1$

$|y(1)| = 1$

6. In an equation of the form : $y\,f(xy)\,dx + xg(xy)\,dy = 0$ the variables can be separated by the substitution $xy = v$. Consider the example **[O]**

$(x^3y^3 + x^2y^2 + xy + 1)\,y\,dx + (x^3y^3 - x^2y^2 - xy + 1)\,x\,dy = 0$

Ans. $2\ell nx + xy + 2\ell n(xy) + \frac{1}{xy} = c$

Sol. $xy = v \Rightarrow y = \frac{v}{x}$

$$x\frac{dy}{dx} + y = \frac{dv}{dx} \quad ....(1)$$

now equation become $y.(v^3 + v^2 + v + 1) + (v^3 - v^2 - v + 1)\,x.\,\frac{dy}{dx} = 0$

$$y.(v^3 + v^2 + v + 1) + (v^3 - v^2 - v + 1)\left[\frac{dv}{dx} - y\right] = 0$$

$$y.[\,(v^3 + v^2 + v + 1) - (v^3 - v^2 - v + 1)\,] + \frac{dv}{dx}(v^3 - v^2 - v + 1) = 0$$

$$\frac{v}{x}(2v^2 + 2v) + \frac{dv}{dx}(v^3 - v^2 - v + 1) = 0$$

$$\frac{2v^2(v+1)}{x} + \frac{dv}{dx}(v^3 - v^2 - v + 1) = 0 \Rightarrow \quad 2\int\frac{dx}{x} + \int\frac{v^3 - v^2 - v + 1}{v^2(v+1)}dv$$

$$2\int\frac{dx}{x} + \int\frac{(v^2-1)(v-1)}{v^2(v+1)}dv \quad \text{or} \quad 2\int\frac{dx}{x} + \int\frac{v^2 - 2v + 1}{v^2}dv\ ]$$

7. **Series of substitution**

$(x\cos y - y\sin y)\,dy + (x\sin y + y\cos y)\,dx = 0$

[Sol. Let $x\,\text{siny} + y\,\text{cosy} = t$

diff. w.r.t. x $\quad x\cos y\frac{dy}{dx} + \sin y + \cos y\frac{dy}{dx} - y\sin y\frac{dy}{dx} = \frac{dt}{dx}$

$$(x\cos y - y\sin y)\frac{dy}{dx} + \sin y\cos y\frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - (\sin y + \cos y\frac{dy}{dx}) + t = 0$$

$$\sin y = z \;;\; \cos y\frac{dy}{dx} = \frac{dz}{dx}$$

$$\frac{dt}{dx} - \left(z + \frac{dz}{dx}\right) + t = 0 \qquad \text{or} \qquad \frac{d}{dx}(t - z) + (t - z) = 0$$

Let $t - z = u$

$$\frac{du}{dx} + u = 0 \quad \Rightarrow \quad \ln u + x = C \quad \Rightarrow \quad \ln(t - z) + x = C$$

$\ln(x\sin y + y\cos y - \sin y) = C - x \Rightarrow x\sin y + y\cos y - \sin y = C\,e^{-x}$]